Homework 5

Friday, April 24, 2026 1:18 PM

6A.2

(a) $V = \frac{Q}{A} = \frac{18\ \text{gal/min}}{3.068\ \text{in}} = \frac{0.0401\ \text{ft}^3/\text{s}}{0.2557\ \text{ft}} = 0.781\ \text{ft/s}$

$\Delta z = 50 \sin(45)^\circ = 35.36\ \text{ft}$

$Re = 1.85 \times 10^4$

$f_D = 0.0277$

$h_f = (0.0277)\left(\frac{95}{0.2557}\right)\left(\frac{0.781^2}{2(32.2)}\right) = 0.098\ \text{ft}$

$\frac{P}{\gamma} = 35.36 + 0.098 = 35.46\ \text{ft}$

$P = (35.46)(0.433) = 15.4\ \text{psi}$

(b) $\frac{0.098}{35.46} \times 100 = 0.28\%$

6A.3

$h_f = \frac{\Delta P}{\gamma} = \frac{36}{62.3} = 0.578\ \text{ft}$

$h_f = f \frac{L}{D} \frac{v^2}{2g}$

$v = \sqrt{\frac{2gh_f D}{fL}}$

$f = 0.079\, Re^{-0.25}, \quad Re = \frac{\rho v D}{\mu}$

$v = 0.915\ \text{ft/s}$

$Re = 1.37 \times 10^6$

$Q = vA = v\left(\frac{\pi D^2}{4}\right) = 0.915 \times \frac{\pi (0.5)^2}{4} = 0.1796\ \text{ft}^3/\text{s} = 4.84 \times 10^3\ \text{gal/hr}$

6A.9

$P_1^2 - P_2^2 = \frac{2RTLG}{M}\left[\frac{150(1-\varepsilon)^2 \mu}{D_p^2 \varepsilon^3} + \frac{1.75(1-\varepsilon)G}{D_p \varepsilon^3}\right]$

$A = \frac{\pi D^2}{4} = 81.07\ \text{cm}^2$

$M = 44.01\ \text{g/mol}$

$G = 5.92\ \frac{\text{g}}{\text{cm}^2 \cdot \text{s}}$

$\dot{m} = 5.92 \cdot 81.07 = 480\ \text{g/s}$

6B.2

(a) $F_k = 1.328 \rho v_\infty^2 W L Re_L^{-1/2}$

$f = \frac{1.328 \rho v_\infty^2 W L Re_L^{-1/2}}{\rho v_\infty^2 W L}$

$f = 1.328\, Re_L^{-1/2}$

(b) $f = \frac{F_k}{\frac{1}{2}\rho v_\infty^2 (2WL)}$

$f = 0.072\, Re_L^{-1/5}$

6B.3

$\langle v_z \rangle = \frac{B^2}{3\mu}\left(\frac{\Delta P}{L}\right)$

$\Delta P = \frac{3\mu L \langle v_z \rangle}{B^2}$

$f = \frac{\Delta P}{\frac{1}{2}\rho \langle v_z \rangle^2} \cdot \frac{B}{L}$

$f = \frac{3\mu L \langle v_z \rangle}{B^2} \cdot \frac{1}{\frac{1}{2}\rho L \langle v_z \rangle^2} \cdot \frac{B}{L}$

$f = \frac{6\mu}{B \rho \langle v_z \rangle}$

$Re = \frac{2B \langle v_z \rangle \rho}{\mu}$

$f = \frac{12}{Re}$

6B.5

(a) $\Delta P = f \frac{L}{D} \frac{\rho v^2}{2}$

$w = \rho V A = \rho v \frac{\pi D^2}{4}$

$v^2 = \frac{1}{D^4}$

$f \propto Re^{-1/4}$

$Re = \frac{\rho v D}{\mu} \propto \frac{1}{D}$

$D_{new} = \frac{D}{2}$

$\frac{\Delta P_{new}}{\Delta P_{old}} = \left(\frac{D}{D/2}\right)^{19/4} = 2^{19/4}$

$2^{19/4} = 27$

$\Delta P_{new} = 27 \cdot \Delta P_{old}$

(b) $\frac{\Delta P_{new}}{\Delta P_{old}} = \frac{f_{new}}{f_{old}} \cdot \left(\frac{D_{old}}{D_{new}}\right)^5 = \frac{f_{new}}{f_{old}} \cdot 32$

6B.7

(a) $m \frac{dv}{dt} = mg - kv^2$

$\frac{dv}{dt} = g - c^2 g v^2$

$\frac{dv}{dt} = g(1 - c^2 v^2)$

$\frac{dv}{1 - c^2 v^2} = g\, dt$

$v = \frac{1}{c} \tanh(cgt)$

$z = \frac{1}{c^2 g} \ln(\cosh(cgt))$

(b) $0 = g(1 - c^2 v_t^2)$

$c^2 v_t^2 = 1$

$v_t = \frac{1}{c}$